

Econometrics

Final examination

Monday, March 16, 2020, 10:00 - 13:30

Instructor: Dr. Doron Israeli

Instructions

You have 3.5 hours to solve this exam.

Work on this exam is to be done individually.

You may use your notes and the R examples presented in class.

When submitting your solutions, be sure to upload to provide soft copies of:

1. A *.doc file named “**FinalExam_ID.doc**”, which comprises your solutions to relevant questions (ID indicates your identification number). The *.doc file should contain detailed solutions to the questions, including the relevant R code and the output.
2. An R script named “**FinalExam_ID.R**”, which comprises the entire R program you used to solve the questions. The R script should
 - be ready to run as it is and generate the results you report in the *.doc file.
 - be properly structured, e.g., indicate the number of question before each sequence of commands, and provide comments to facilitate understanding.

General

The **data set** needed to answer computer questions in this exam are available in the **ISLR** package of the computation and visualization software R.

Recall, in order to access a data set available in an R package, you need to

- (1) install the package,
- (2) load the package into R, and
- (3) load the desired data set.

After completing these steps, the desired data set will be available for your use as a named object (in a *data frame* format).

For example, to load the **Auto** data set from the **ISLR** package, you need to implement the following commands:¹

```
install.packages("ISLR")
require(ISLR)
data(Auto)
```

To make sure that **all functionality** needed to solve this exam are properly installed and loaded into R, **run the following commands, before solving the exam questions:**

```
options(digits=3, scipen=10)
```

```
install.packages("lmtest")
install.packages("zoo")
require(lmtest)
```

```
install.pacakges("moments")
require(moments)
```

```
install.packages("tseries")
require(tseries)
```

```
install.packages("sandwich")
require(sandwich)
```

```
install.packages("car")
require(car)
```

```
install.packages("ISLR")
require(ISLR)
data(Auto)
```

¹ Recall, you need to install the package only once.

Questions

This question uses the **Auto** data set (from the **ISLR** package), which comprises 392 observations (for 392 vehicles) on the following nine variables.

mpg	miles per gallon
cylinders	number of cylinders (between 4 and 8)
displacement	engine displacement (cu. inches)
horsepower	engine horsepower
weight	vehicle weight (lbs.)
acceleration	time (sec.) to accelerate from 0 to 60 mph
year	model year (modulo 100)
origin	origin of car (1 for American; 2 for European, and 3 for Japanese)
name	vehicle name

Suppose that you are interested in exploring how various car specifications affect its fuel economy. Based on your intuition, you postulate the following multiple linear regression (MLR) model (Eq. 1):

$$mpg = \beta_0 + \beta_1 cylinders + \beta_2 displacement + \beta_3 horsepower + \beta_4 weight + U.$$

Prior to estimating Eq. 1 by OLS you met Prof. Israeli to get his input. Prof. Israeli was in a rush and quickly told you that prior research suggests that, all else being equal:

- the relation between *displacement* and *mpg* depends on the number of cylinders, *cylinders*, of the car;
- *horsepower* has a quadratic relation with *mpg*, and
- the relation between *weight* and *mpg* has a diminishing marginal effect (i.e., a % change in *weight* has a constant effect on *mpg*).

(1) (8 points) Write down the MLR model that Prof. Israeli has in mind (Eq. 2). Explain the changes you make to Eq. 1 when constructing Eq. 2.

(2) (8 points) Estimate Eq. 1 and Eq. 2 by OLS. Report and interpret the coefficient estimates in each Equation.² In your interpretations, be sure to identify and discuss the estimated slope of the relation between the expected value of *mpg* and each of the following three explanatory variables: *displacement*, *horsepower*, and *weight*.

(3) (10 points) What is the estimated partial effect on the expected value of *mpg* of

- *displacement*, for a car with a sample average number of *cylinders*?
- *horsepower*, for a car with a sample average number of *horsepower*?
- *weight*?

Be sure to provide two sets of estimates, one for Eq. 1 and one for Eq. 2.

² Focus on coefficient estimates and not on their significance.

(4) (5 points) Which MLR model (Eq. 1 or Eq. 2) better explains variation in a car's fuel economy? Explain your answer.

(5) (20 points) Using a significance level of 5% and the estimates obtained from Eq. 2, test the following hypotheses for the population of American; European, and Japanese cars:

- **(10 points)** a 1% increase in *weight* reduces the average miles per gallon by 0.1;
- **(10 points)** the effect of a one unit increase in *displacement* on the average miles per gallon for a car with 4 *cylinders* is equal to the estimate you found in part (3) for a car with a sample average number of *cylinders*.

Be sure to use the appropriate standard errors when conducting the tests. Explain your assumptions and computations.³

(6) (18 points + 5 bonus points) In part (5) you relied on the classical approach to statistical inference. In particular, you assumed that either the errors in Eq. 2 follow the normal distribution or the n is large enough to rely on large sample approximations. As we discussed in the classroom, such a naïve approach can be proven wrong in various situations.

- **(6 points)** Conduct a non-formal test for whether the errors in Eq. 2 are normally distributed. What do you learn from the test? Explain.
- **(6 points)** Using a significance level of 5%, conduct a formal test for whether the errors in Eq. 2 are normally distributed. What do you learn from the test? Explain.
- **(6 points)** Given your conclusions from the non-formal as well as formal tests above, express your opinion on inferences you derived in part (5).
- **(5 bonus points)** Using “Bootstrapping the sample” approach for MLR, find a two-tailed 95% Bootstrap interval estimate for the coefficient in Eq. 2 that you used to find the partial effect of *weight* on the expected value of *mpg*. What do you learn from this interval estimate? In your answer be sure to refer to the hypothesis test you conducted in part (3).
 - Base the Bootstrap sampling distribution on 10,000 resamples.
 - Use the standard error method for obtaining the interval estimate.

(7) (10 points) Suppose that shortly after the meeting, Prof. Israeli sent you a message in which he expressed his disappointment. Prof. Israeli wrote that he could not understand what intuition brought you to postulate the model in Eq. 1. In search for an appropriate excuse, you decided to tell your professor that the dependent variable that you had in mind is a binary version of *mpg*, *mpg_b*, and not as it appears in the *Auto* data set. Thus, you have not considered the types of relations among variables that he suggested during the meeting.

³ Use the classical approach to statistical inference (i.e., assume normality of the errors or a large enough sample size to invoke the Central Limit Theorem, CLT).

- Create a dummy variable, *mpg_b*, which takes on the value of 1 if *mpg* is in the top 25% of the sample distribution of *mpg* and 0 otherwise.
- Estimate Eq. 1 by OLS but use *mpg_b* (not *mpg*) as the dependent variable. Interpret the coefficient estimates you get and comment on the type of MLR model that you estimate.

(8) (21 points) It seems like Prof. Israeli did not like your response. Immediately after receiving your message, Prof. called you and, with anger, asked you: Did you estimate the new version of Eq. 1 (i.e., with *mpg_b* as the dependent variable) by OLS? It does not make any sense! To this you responded: Prof. Israeli, of course not! I used the Logit model!

- **(5 points)** What did Prof. Israeli have in mind when he said "It does not make any sense?"
- **(6 points)** Estimate the Logit model of the new version of Eq. 1 (i.e., with *mpg_b* as the dependent variable) by MLE. Interpret the coefficient estimates you get and compare them to those you obtained in part (7).
- **(10 points)** What is the average partial effect (APE) of *weight* on the probability that the level of a car's miles per gallon falls in the top 25% of the sample distribution of *mpg*.⁴

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⁴ Assume that that *weight* is a roughly continuous variable.